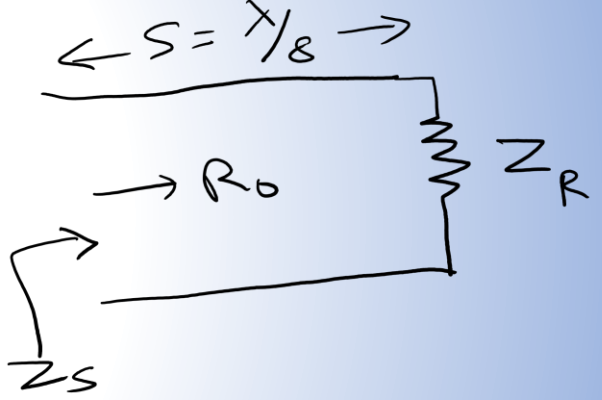
Eighth, quarter and half wave lines

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The Eighth wave line:

- Any line whose length is s = λ/8 is called eighth wave line

$$Z_S = R_0 \frac{Z_R + j R_0 \tan \beta s}{R_0 + j Z_R \tan \beta s}$$



- If s = λ/8, βs = (2π/λ)(λ/8) = π/4

$$Z_s = R_0 \left[\frac{Z_R + jR_0 \tan (\pi/4)}{R_0 + jZ_R \tan (\pi/4)} \right]$$

$$= R_0 \left(\frac{Z_R + jR_0}{R_0 + jZ_R} \right)$$

- If such a line is terminated with a pure resistance R_R

$$Z_s = R_0 \left(\frac{R_R + jR_0}{R_0 + jR_R} \right)$$

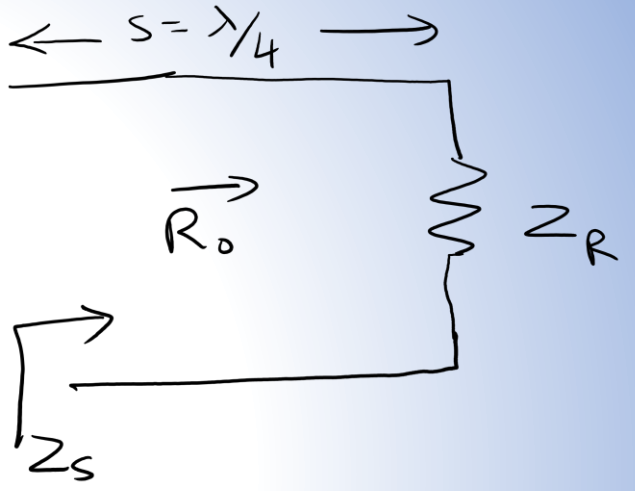
$$|Z_s| = R_0 \frac{\sqrt{R_R^2 + R_0^2}}{\sqrt{R_0^2 + R_R^2}}$$

$$|Z_s| = R_0$$

- Thus an eighth-wave line may be used to obtain a magnitude match between a resistance of any value and a source of R_0 initial resistance.

The Quarter- wave line:

$$Z_s = R_0 \frac{Z_R + j R_0 \tan \beta s}{R_0 + j Z_R \tan \beta s}$$



- If $s = \lambda/4$, $\beta s = (2\pi/\lambda)(\lambda/4) = \pi/2$

$\tan(\pi/2) = \infty$

$$Z_s = R_0 \left[\frac{\frac{Z_R}{\tan(2\pi s/\lambda)} + jR_0}{\frac{R_0}{\tan(2\pi s/\lambda)} + jZ_R} \right]$$

$$Z_s = \frac{R_0^2}{Z_R}$$

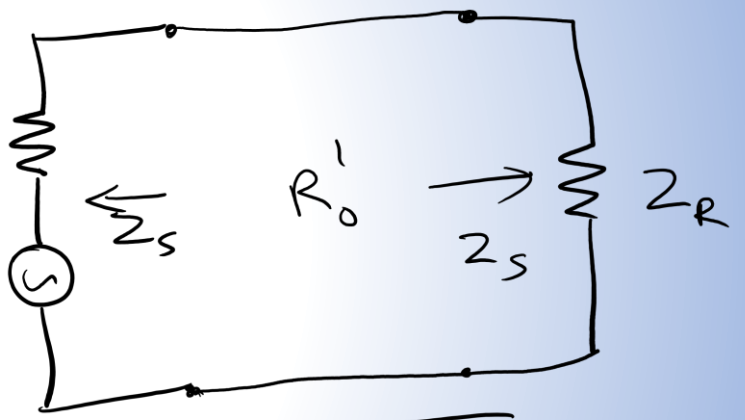
$$R_0 = |\sqrt{Z_s Z_R}|$$

Application 1:

A quarter wave selection of a line may be thought of as a transformer to match a load of Z_R Ω to a source of Z_S Ω

Eg: $Z_S = 50\,\Omega$, $Z_R = 100\,\Omega$

$$R_0' = \sqrt{50 \times 100} = 70.71$$



$$R_0' = \sqrt{Z_S Z_R}$$

For flat cable

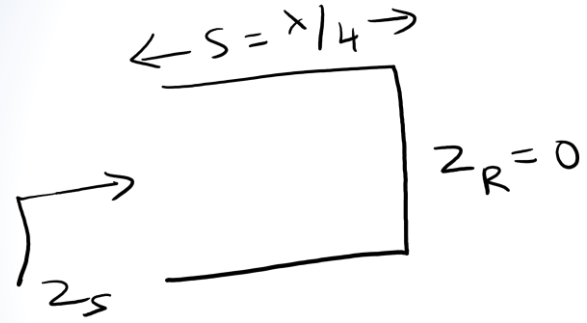
$$70.71 = \frac{120}{\sqrt{\epsilon_r}} \ln\left(\frac{d}{a}\right)$$

For co-axial cable

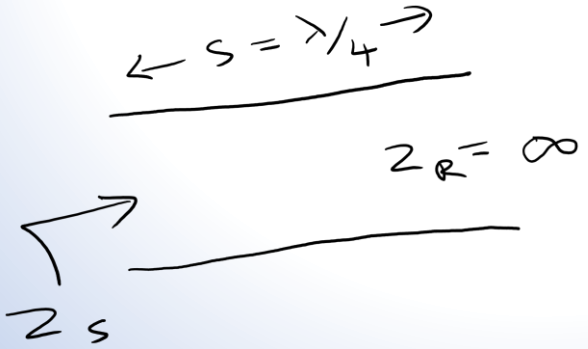
$$70.71 = \frac{60}{\sqrt{\epsilon_r}} \ln\left(\frac{b}{a}\right)$$

Application 2:

- A quarter-wave line may be considered as an impedance inverter
- It can transform a low impedance into high impedance and vice versa



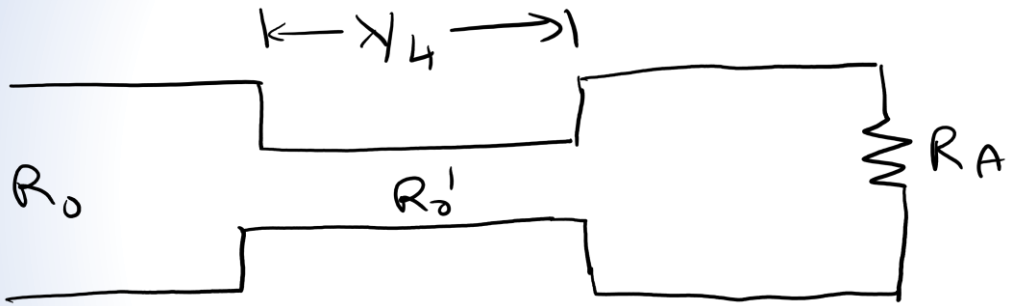
$$Z_{sc} = \frac{R_0^2}{Z_R} = \infty$$



$$Z_{oc} = \frac{R_0^2}{Z_R} = 0$$

Application 3:

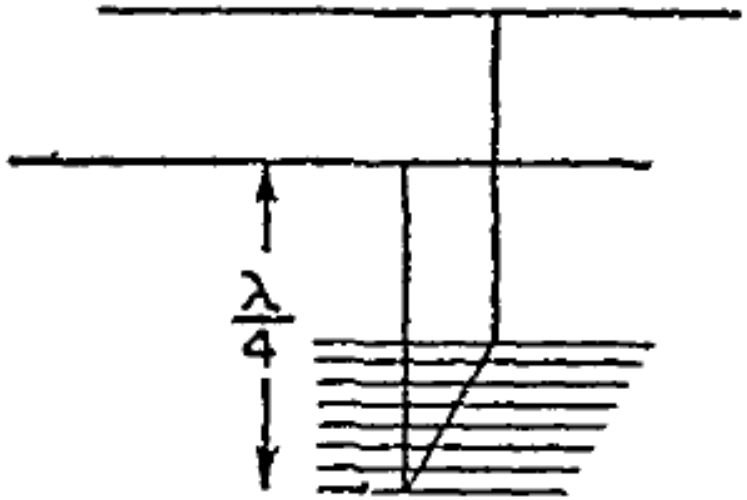
- To couple a transmission line to a resistive load such as antenna where $R_0' = \sqrt{R_0 R_A}$



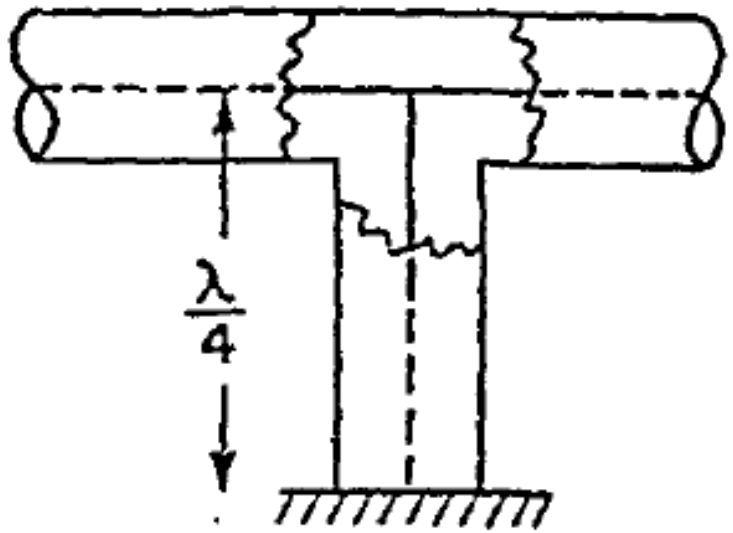
- If physical spacing required is greater than quarter wave length, odd number of wave lines can be used
- The range of values of R_A and R_0 that can be matched satisfactorily is limited - roughly about 10 to 1

Application 4: Quarter - wave lines as insulators

Open-wire line

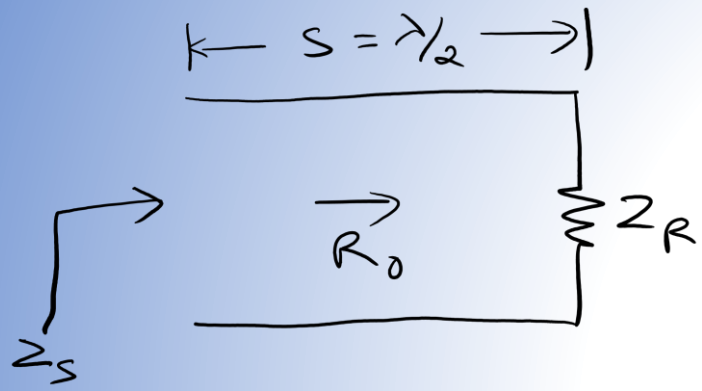


Co-axial cable



- Figure shows – insulator to support open wire line or the center conductor of coaxial cable – a quarter wave line is used
- The input impedance of a quarter-wave shorted line is ∞
- They are some times referred to as copper insulators

The half-wave line:



$$\beta s = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{2} = \pi$$

$$\tan \beta s = \tan \pi = 0$$

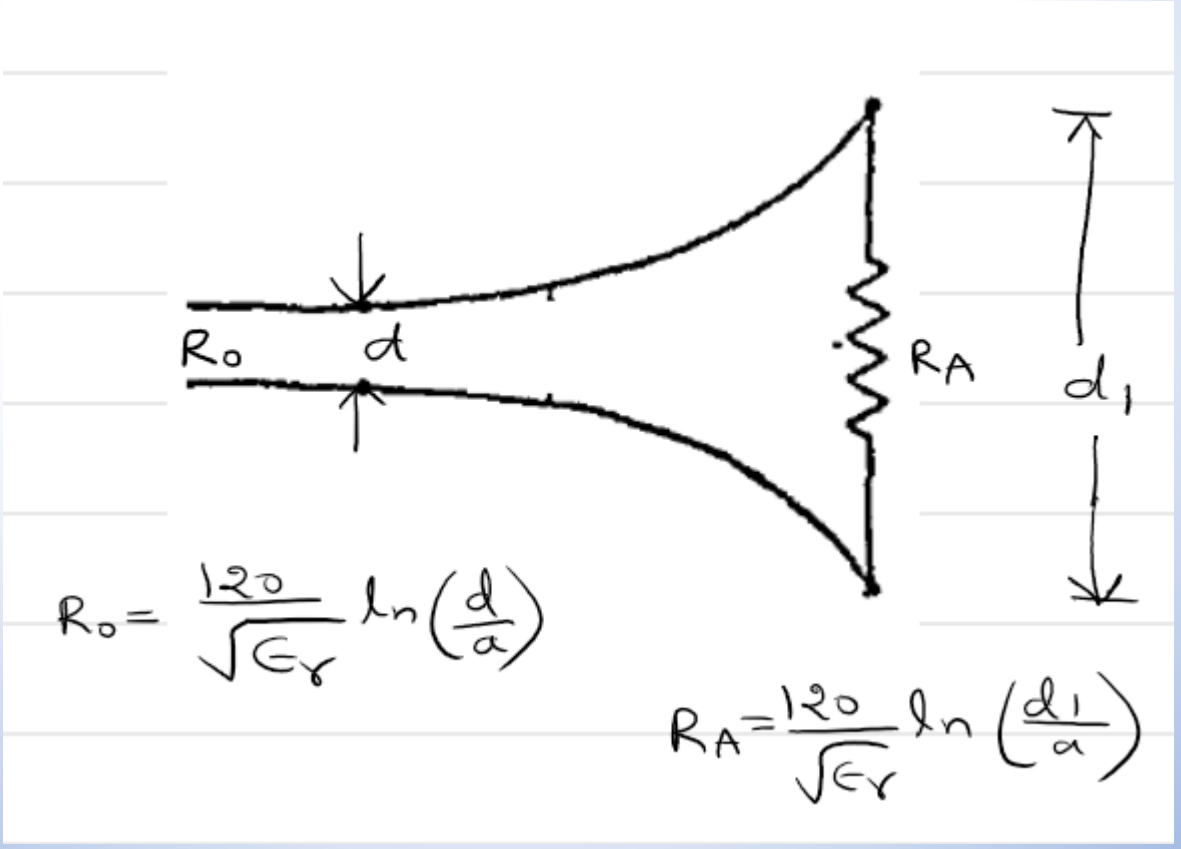
$$Z_S = R_0 \left(\frac{Z_R + j R_0 \tan \pi}{R_0 + j Z_R \tan \pi} \right)$$

$$\boxed{Z_S = Z_R}$$

- It is considered as one-to-one (1 : 1) transformer
- It has its greatest utility in connecting a load to a source in cases where the load and source can’t be made adjacent

Exponential line (mainly used for impedance matching)

- The characteristic impedance of a line is a function of spacing and size of the conductors among other factors
- If the spacing of a dissipation less line is made to vary in a uniform manner along the length of a line – R_0 will likewise vary along the line




Thank
you